13A 1 EQUILIBRIUM CONSTANT, K_c

SPECIFICATION REFERENCE
13.1 13.3 PART

LEARNING OBJECTIVES

- Be able to deduce an expression for K_c, for homogeneous and heterogeneous systems, in terms of equilibrium concentrations.
- Be able to calculate a value, with units where appropriate, for the equilibrium constant K_c for homogeneous and heterogeneous reactions, from experimental data.

HOMOGENEOUS REACTIONS

A **homogeneous reaction** is one in which all reactants and products are in the same phase. For example, all are gases or all are in aqueous solution.

In **Topic 9 (Book 1: IAS)** we saw how the position of equilibrium of a reversible reaction can change when the conditions are altered.

In this section, we will study *quantitatively* how changes in concentration affect the equilibrium position.

For the reversible reaction:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

it can be shown experimentally that at equilibrium and at a given temperature:

$$\frac{[NO_2(g)]^2_{eq}}{[N_2O_4(g)]_{eq}} = \text{a constant}$$

This value is called the **equilibrium constant** and is given the symbol K_c.

The subscript 'c' indicates that this equilibrium constant is calculated using concentrations. This is to distinguish it from another equilibrium constant, K_p, which uses partial pressures.

The subscript 'eq' indicates that the concentrations are those *at equilibrium*.

At a given temperature, the value of the equilibrium constant does not change with concentration, unlike the equilibrium position. The only factor that affects the value of the equilibrium constant is temperature.

There are two ways of obtaining a value for the equilibrium constant:

- the first is by experimentation, and this is the method we will look at in this section
- the second is by calculation using thermodynamic equations, and this will be covered in **Section 13A.5**.

For a general reaction:

$$wA + xB \rightleftharpoons yC + zD$$

$$K_c = \frac{[C]^y_{eq}[D]^z_{eq}}{[A]^w_{eq}[B]^x_{eq}}$$

For convenience, the 'eq' sign is often not included, but is taken for granted.

DETERMINATION OF THE EQUILIBRIUM CONSTANT

To determine the value of the equilibrium constant for a reaction, known amounts of reactants are allowed to reach equilibrium with their products. At equilibrium, the amount of one of the substances is measured and the amounts of the others can be calculated using the stoichiometry of the equation.

WORKED EXAMPLE 1

2.00 mol of ethanoic acid and 2.00 mol of ethanol are mixed and allowed to reach equilibrium with ethyl ethanoate and water at 298 K. The amount of ethanoic acid at equilibrium is found to be 0.67 mol. Calculate K_c for the reaction at 298 K.

Answer

Work out the amounts, and hence the concentrations, of each substance at equilibrium:

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

amount at start/mol			
2.00	2.00	0	0
amount at equilibrium/mol			
0.67	0.67	(2.00 – 0.67) = 1.33	(2.00 – 0.67) = 1.33
concentration at equilibrium/mol dm^{-3}			
$\frac{0.67}{V}$	$\frac{0.67}{V}$	$\frac{1.33}{V}$	$\frac{1.33}{V}$

where V is the total volume in dm^3 of the reaction mixture.

$$K_c = \frac{[CH_3COOCH_2CH_3(l)][H_2O(l)]}{[CH_3COOH(l)][CH_3CH_2OH(l)]} = \frac{(1.33/V)(1.33/V)}{(0.67/V)(0.67/V)}$$

$$= 3.94 \text{ (no units)}$$

K_c has no units because the same number of moles are present on each side of the equation. For the same reason, V also cancels.

WORKED EXAMPLE 2

0.100 mol of N_2O_4 is placed into a 0.100 dm^3 flask and allowed to reach equilibrium with NO_2 at 398 K. At equilibrium, there is 0.071 mol of N_2O_4. Calculate K_c at this temperature.

Answer

$N_2O_4(g)$	$\rightleftharpoons$	$2NO_2(g)$	
0.100		0	amount at start/mol
0.071		(0.100 – 0.071) × 2 = 0.058	amount at equilibrium/mol
0.071/0.100		0.058/0.100	concentration at equilibrium/mol dm^{-3}
= 0.71		= 0.58	

$$K_c = \frac{[NO_2(g)]^2}{[N_2O_4(g)]} = \frac{0.58^2}{0.71} = 0.474 \text{ mol dm}^{-3}$$

Note: this time, K_c has units of mol dm^{-3}

$$\frac{\text{mol dm}^{-3} \times \text{mol dm}^{-3}}{\text{mol dm}^{-3}} = \text{mol dm}^{-3}$$

HETEROGENEOUS REACTIONS

A **heterogeneous reaction** is one in which at least one of the reactants and/or products is in a different phase to the others.

Examples include:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

$$3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g)$$

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

WRITING EQUILIBRIUM CONSTANTS FOR HETEROGENEOUS REACTIONS

If we apply the equilibrium law to the reaction:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

we obtain:

$$\frac{[CaO(s)][CO_2(g)]}{[CaCO_3(s)]} = \text{a constant}$$

DID YOU KNOW?
The concentration term of a solid in a heterogeneous equilibrium is constant. It can therefore be absorbed into the equilibrium constant.

However, the concentration of a solid at a given temperature is determined by its density, which has a constant value. Hence, the expression can be simplified to:

$[CO_2(g)]$ = a constant

This constant is given the symbol K_c.

So, $K_c = [CO_2(g)]$ for the above equilibrium.

Similarly, for the equilibrium:

$$3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g)$$

$$K_c = \frac{[H_2(g)]^4}{[H_2O(g)]^4}$$

CHECKPOINT

1. A dynamic equilibrium is set up between carbon monoxide, hydrogen and methanol. The equation for the reaction is:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

The equilibrium concentrations, in mol dm^{-3}, for the three components are shown below:

$CO(g)$, 3.1×10^{-3}; $H_2(g)$, 2.4×10^{-2}; $CH_3OH(g)$, 2.6×10^{-5}

(a) Write an expression for K_c for this reaction.

(b) Calculate a value for K_c. State the units, if any.

SKILLS ADAPTIVE LEARNING

2. The reaction between ethanoic acid and ethanol is reversible. The equation for the reaction is:

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

6.0 mol of ethanoic acid and 12.5 mol of ethanol are mixed together with some sulfuric acid to act as a catalyst. The total volume of the reaction mixture is V dm^3. The mixture is left for several days to reach equilibrium, after which only 1.0 mol of ethanoic acid remains.

(a) Write an expression for Kc for the reaction.

(b) Calculate the amounts of ethanol, ethyl ethanoate and water present in the equilibrium mixture.

(c) Use your answers to part (b) to calculate a value for Kc. Give your answer to an appropriate number of significant figures and state the units, if any.

3. The reaction between hydrogen and iodine to form hydrogen iodide is reversible.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

0.30 mol of $H_2(g)$ is mixed with 0.20 mol of $I_2(g)$ and the mixture is allowed to reach equilibrium. At equilibrium, 0.14 mol of $H_2(g)$ is present.

Calculate the value of K_c for the reaction. Give your answer to an appropriate number of significant figures and state the units, if any.

4. Nitrogen and oxygen can exist in equilibrium with nitrogen(II) oxide. The equation for the reaction is:

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g) \qquad \Delta H \text{ is positive}$$

The equilibrium constant at 289 K for this reaction is 4.8×10^{-31}.

(a) What does the value of the equilibrium constant tell you about the position of equilibrium of this reaction at 298 K?

(b) An equilibrium mixture of volume 1.2 dm^3 of the three gases contains 1.1 mol of $N_2(g)$ and 4.0×10^{-16} mol of $NO(g)$. Calculate the equilibrium concentration of $O_2(g)$ in this mixture.

5. The equilibrium constant, K_c, for the reaction:

$$H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g)$$

has a value of 4.92×10^{-5} mol dm^{-3} at 700 K.

A mixture of water vapour and carbon is heated to 700 K in a closed vessel and allowed to reach equilibrium with hydrogen and carbon monoxide.

Write an expression for K_c and use it to calculate the equilibrium concentrations of H_2 and CO in the above equilibrium mixture, at 700 K, when the equilibrium concentration of H_2O is 2.00×10^{-2} mol dm^{-3}.

EXAM HINT

In question 2, the amount of acid at equilibrium can be quantified by titration. You might expect the equilibrium to shift to the left as the acid is neutralised. However, the rate of this reaction is very slow (note that the reaction is left for several days). This means that the amount of acid can be quantified before the equilibrium has had time to respond.

SUBJECT VOCABULARY

homogeneous reaction a reaction in which all the reactants and products are in the same phase

equilibrium constant a number that expresses the relationship between the amounts of products and reactants present at equilibrium in a reversible chemical reaction at a given temperature

heterogeneous reaction a reaction in which at least one of the reactants and/or products is in a different phase to the others

13A 2 EQUILIBRIUM CONSTANT, K_p

SPECIFICATION REFERENCE

13.2 | 13.3 PART

LEARNING OBJECTIVES

- Be able to deduce an expression for K_p for homogeneous and heterogeneous systems in terms of equilibrium partial pressures in atm.
- Be able to calculate a value, with units where appropriate, for the equilibrium constant K_p for homogeneous and heterogeneous reactions from experimental data.

HOMOGENEOUS REACTIONS

For reversible reactions involving gases we can express the concentrations of the reactants and products in terms of their partial pressures. The **partial pressure** of an individual gas in a mixture of gases is defined as the pressure that the gas would exert if it alone occupied the volume of the mixture. The total pressure of the mixture is then equal to the sum of the partial pressures of all the gases present in the mixture.

If we have a mixture of two gases, A and B, then the total pressure, p, of the mixture is related to the partial pressures of A and B by the following equation:

$$p = p_A + p_B$$

where p_A = partial pressure of A and p_B = partial pressure of B.

For the general reaction:

$$wA(g) + xB(g) \rightleftharpoons yC(g) + zD(g)$$

we can define another equilibrium constant, K_p, in terms of partial pressures:

$$K_p = \frac{(p_C)^y(p_D)^z}{(p_A)^w(p_B)^x}$$

CALCULATING PARTIAL PRESSURES

The partial pressure of an individual gas in a mixture of gases is calculated by multiplying its mole fraction by the total pressure, for example:

$$p_A = x_A p$$

where x_A = the mole fraction of gas A in the mixture.

$$\text{mole fraction of A} = \frac{\text{number of moles of A}}{\text{total number of moles of gas}}$$

WORKED EXAMPLE

1.00 mol of PCl_5 vapour is heated to 500 K in a sealed vessel. The equilibrium mixture, at a pressure of 6 atm, contains 0.60 mol of chlorine. Calculate K_p for the reaction:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Answer

$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$	Total	
1.00		0		0	1.00	amount at start/mol
1.00 – 0.60		0.60		0.60	1.60	amount at equilibrium/mol
$\frac{0.40}{1.60}$		$\frac{0.60}{1.60}$		$\frac{0.60}{1.60}$		mole fraction at equilibrium
0.25 × 6		0.375 × 6		0.375 × 6		partial pressure/atm

$$K_p = \frac{(p_{PCl_3})(p_{Cl_2})}{(p_{PCl_5})} = \frac{(0.375 \times 6)(0.375 \times 6)}{(0.25 \times 6)} = 3.38 \text{ atm}$$

EXAM HINT

Note that you must use (p) for partial pressures. Do *not* use square brackets, which are for substances measured in mol dm^{-3}.

HETEROGENEOUS REACTIONS

SOLIDS AND GASES

For solids that are in equilibrium with gases, the partial pressure terms of any solids are not included in the expression for K_p. You will recall that K_c has a similar rule.

Consider the reaction:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

The equilibrium constant in terms of partial pressures is given by:

$$K_p = p_{CO_2(g)}$$

This equation suggests that for a particular temperature, the pressure of carbon dioxide in the equilibrium mixture is constant regardless of the masses of calcium carbonate and calcium oxide present. This prediction can be confirmed experimentally.

At 1073 K, the pressure of carbon dioxide in equilibrium with $CaCO_3(s)$ and $CaO(s)$ is 0.25 atm.

Hence, at 1073 K, K_p = 0.25 atm.

For this reaction, K_p increases with increasing temperature. It only exceeds 1 atm at temperatures above 1173 K. So, calcium carbonate must be heated to this temperature for it to decompose at 1 atm of pressure.

DID YOU KNOW?

A similar situation exists for liquids that are in equilibrium with gases. For the equilibrium between liquid water and water vapour:

$$H_2O(l) \rightleftharpoons H_2O(g)$$

$$K_p = p_{H_2O(g)}$$

The partial pressure of water (p_{H_2O}) in equilibrium with its liquid is also known as the 'saturated vapour pressure' of water.

At 298 K, K_p, and hence p_{H_2O}, is equal to 0.03 atm; whereas at 373 K, K_p, and hence p_{H_2O}, is equal to 1.00 atm.

Since the boiling temperature of a liquid is defined as the temperature at which the saturated vapour pressure equals the external pressure, the boiling temperature of water at 1 atm is 373 K (100 °C).

CHECKPOINT

SKILLS PROBLEM-SOLVING

1. One stage in the manufacture of sulfuric acid by the contact process involves the reaction between sulfur dioxide and oxygen to form sulfur trioxide:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

(a) Write an expression for K_p for this reaction.

(b) An equilibrium is set up for this reaction at 700 K.

At this temperature:

- the partial pressure of SO_2 is 0.100 atm
- the partial pressure of O_2 is 0.500 atm
- K_p for the reaction is 3.00×10^4 atm^{-1}

Calculate the partial pressure of SO_3 in this equilibrium mixture and hence determine the percentage of SO_3 present.

2. The reaction between carbon monoxide and chlorine to form phosgene, $COCl_2$, is reversible. The equation for the reaction is:

$$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$$

Some carbon monoxide and chlorine are allowed to react and reach a position of equilibrium. The equilibrium partial pressures of the mixture are:

$CO(g)$, 2.47×10^{-8} atm; $Cl_2(g)$, 2.47×10^{-8} atm; $COCl_2(g)$, 4.08×10^{-10} atm

(a) What is meant by the term 'partial pressure'?

(b) Write an expression for K_p for this reaction and calculate its value, giving the units, if any.

3. When chlorine gas is heated to a high temperature, the molecules dissociate into chlorine atoms.

$$Cl_2(g) \rightleftharpoons 2Cl(g)$$

Some chlorine gas is placed in a closed container and heated to 1400 K until equilibrium is established. The partial pressure of $Cl_2(g)$ is 0.84 atm and that of $Cl(g)$ is 0.030 atm.

(a) Determine the mole fraction of Cl in the equilibrium mixture.

(b) (i) Write an expression for K_p for this reaction.

(ii) Calculate the value of K_p for this reaction at 700 K. Give the units, if any.

SUBJECT VOCABULARY

partial pressure (of a gas in a mixture of gases) the pressure that the gas would exert if it alone occupied the volume of the mixture

13A 3 FACTORS AFFECTING EQUILIBRIUM CONSTANTS 1

SPECIFICATION REFERENCE 13.6 13.7

LEARNING OBJECTIVES

- Know the effect of changing the temperature on the equilibrium constant (K_c and K_p) for both exothermic and endothermic reactions.
- Understand that the effect of temperature on the position of equilibrium is explained using a change in the value of the equilibrium constant.

EFFECT OF TEMPERATURE ON K_P AND K_C

In this section we will discuss the effects of temperature on equilibrium constants. **Table A** gives the values of the equilibrium constant, K_p, at different temperatures for two gaseous reactions.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $\Delta H^\ominus = -92.2\,kJ\,mol^{-1}$		$N_2O_4(g) \rightleftharpoons 2NO_2(g)$; $\Delta H^\ominus = +57.2\,kJ\,mol^{-1}$	
T/K	K_p/atm^{-2}	T/K	K_p /atm
298	6.76×10^5	298	1.15×10^{-1}
400	4.07×10^1	400	4.79×10^1
500	3.55×10^{-2}	500	1.70×10^3
600	1.66×10^{-3}	600	1.78×10^4

table A

For the *exothermic* reaction between $N_2(g)$, $H_2(g)$ and $NH_3(g)$, the value of K_p *decreases* with increasing temperature.

The opposite is true for the *endothermic* reaction between $N_2O_4(g)$ and $NO_2(g)$.

Exactly the same trend is observed with the values of K_c.

Table B summarises the effect of changing the temperature on the value of the equilibrium constant.

THERMICITY OF REACTION	INCREASE IN TEMPERATURE	DECREASE IN TEMPERATURE
exothermic	K decreases	K increases
endothermic	K increases	K decreases

table B

CHANGE IN TEMPERATURE ALTERS THE EQUILIBRIUM POSITION

If the equilibrium constant for a reaction changes with a change in temperature, then it follows that the equilibrium position also changes. This is illustrated by the data in **table C**, which shows the value of K_c and the amounts of reactants and products at equilibrium for the reaction between hydrogen and iodine to form hydrogen iodide:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad \Delta H^\ominus = -9\,kJ\,mol^{-1}$$

In each case, one mole of hydrogen is mixed with one mole of iodine and equilibrium is allowed to establish, at the given temperature, in a vessel of volume 1 dm^3.

TEMPERATURE	K_p	AMOUNT/mol		
		$H_2(g)$	$I_2(g)$	$HI(g)$
500	160	0.14	0.14	1.72
700	54	0.21	0.21	1.58

table C

As the value of K_p decreases, the percentages of $H_2(g)$ and $I_2(g)$ increase, and the percentage of HI(g) decreases. The equilibrium shifts to the left, in the endothermic direction, as the temperature is increased.

CHECKPOINT

1. A dynamic equilibrium is set up between carbon monoxide, hydrogen and methanol. The equation for the reaction is:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

When the reaction is carried out at a higher temperature, the value of K_c decreases.

Explain whether the forward reaction is exothermic or endothermic.

SKILLS ADAPTIVE LEARNING

2. The reaction between hydrogen and iodine to form hydrogen iodide is reversible:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Some $H_2(g)$ is mixed with $I_2(g)$ and the mixture is allowed to reach equilibrium. At equilibrium, 0.14 mol of $H_2(g)$ is present.

The experiment is repeated with the same amounts of $H_2(g)$ and $I_2(g)$, but at a higher temperature. This time the amount of $H_2(g)$ at equilibrium is greater than 0.14 mol.

Explain what this information tells you about the reaction.

3. Nitrogen and oxygen can exist in equilibrium with nitrogen(II) oxide. The equation for the reaction is:

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g) \qquad \Delta H \text{ is positive}$$

The equilibrium constant at 298 K for this reaction is 4.8×10^{-31}.

Explain the change, if any, in the proportion of NO(g) at a temperature higher than 298 K.

13A 4 FACTORS AFFECTING EQUILIBRIUM CONSTANTS 2

SPECIFICATION REFERENCE 13.4 13.5

LEARNING OBJECTIVES

- Understand how, if at all, a change in temperature, pressure or the presence of a catalyst affects the equilibrium composition in a homogeneous or heterogeneous system.
- Understand that the value of the equilibrium constant is not affected by changes in concentration or pressure or by the addition of a catalyst.

EFFECT OF CONCENTRATION

In **Topic 9 (Book 1: IAS)** we saw that the equilibrium position changes when the concentration of a component of the equilibrium mixture is altered. In this section we will discuss the effects of concentration, pressure and catalysts on equilibrium constants and provide an explanation for this phenomenon in terms of K_c.

The most important thing to appreciate is that the only factor that can change the value of K_c is a change in temperature. That is, *K_c is constant at a given temperature.*

When the concentration of one of the components of an equilibrium mixture is altered, there is an immediate change in the **reaction quotient**, Q_c, which is the mathematical relationship between the concentrations of the components.

At this point, the reaction quotient is no longer equal to the equilibrium constant, K_c. Therefore, the equilibrium composition changes until the reaction quotient and equilibrium constant become equal.

The following examples illustrate this process.

REACTION QUOTIENT, Q_c

For a general reaction:

$$wA + xB \rightleftharpoons yC + zD$$

the reaction quotient, Q_c, is given by the expression:

$$Q_c = \frac{[C]^y[D]^z}{[A]^w[B]^x}$$

where the concentration terms are not necessarily those at equilibrium.

At equilibrium $Q_c = K_c$.

WORKED EXAMPLE 1

The equation for the reaction between hydrogen and iodine to form hydrogen iodide is:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Explain the effect on the position of equilibrium of increasing the concentration of hydrogen at a constant temperature and constant volume.

$$K_c = \frac{[HI(g)]^2_{eq}}{[H_2(g)]_{eq}[I_2(g)]_{eq}} \qquad Q_c = \frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$$

- If $[H_2(g)]$ is suddenly increased, then Q_c decreases and is no longer equal to K_c .
- In order for equilibrium to be re-established, the magnitude of the denominator in the expression for Q_c has to decrease in value.
- This will produce a subsequent increase in the value of the numerator.
- The two values will adjust until the value of Q_c becomes equal to that of K_c and a new equilibrium system is established.

The net result of an increase in $[H_2(g)]$ is that the equilibrium shifts to the right. This is in agreement with the qualitative prediction that we made in **Topic 9 (Book 1: IAS)**.

QUALITATIVE PREDICTIONS

In **Topic 9 (Book 1: IAS)** we said that qualitative predictions do not always give us the correct result. The worked example below shows this.

WORKED EXAMPLE 2

The amounts of N_2, H_2 and NH_3 in an equilibrium mixture are 0.510 mol, 0.197 mol and 0.204 mol, respectively:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The total volume of the gaseous mixture is 1.00 dm^3.

Work out the direction of the shift in equilibrium position if 0.140 mol of N_2 gas is suddenly added to the equilibrium system at constant temperature and pressure.

Show your calculations.

$$Q_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$= \frac{(n_{NH_3}/V)^2}{(n_{N_2}/V)(n_{H_2}/V)^3}$$

where n is the amount (in mol) and V is the total volume (in dm^3) of the mixture.

(Remember that concentration = amount ÷ volume.)

Step 1: Calculation of K_c.

$$K_c = \frac{(0.204)^2/1^2}{(0.510)/1 \times (0.197)^3/1^3} = 10.7$$

Step 2: Calculation of new total number of moles of gas.

Original total moles of gas = (0.510 + 0.197 + 0.204) = 0.911

New total moles of gas = (0.911 + 0.140) = 1.051

Step 3: Calculation of new volume of gas mixture.

Since the pressure remains constant, the gas mixture will expand. If we assume that the gas mixture behaves like an ideal gas, then the volume of gas is proportional to the number of moles:

$$n_1/V_1 = n_2/V_2$$

So the new volume of the gas mixture =

$(1.051 \div 0.911) \times 1\ dm^3 = 1.154\ dm^3$

Step 4: Calculation of Q_c immediately after the addition of nitrogen.

$$Q_c = \frac{(0.204)^2/(1.154)^2}{(0.650)/1.154 \times (0.197)^3/(1.154)^3} = 11.15$$

Step 5: Compare Q_c with K_c.

$11.15 > 10.7$, therefore $Q_c > K_c$.

For equilibrium to be re-established, Q_c must decrease. This can only happen by the equilibrium shifting to the left, to increase the denominator in the equation.

The addition of more nitrogen, at constant pressure, results in a shift of the equilibrium position to the left. Our prediction told us that the equilibrium would shift to the right, away from the nitrogen added.

The problem with making predictions is that two conflicting changes are occurring here:

1. the increase in the number of moles of nitrogen should result in a shift of equilibrium to the right
2. the increase in volume should result in a shift to the left.

We have no way of knowing which effect is greater unless we perform the calculations shown above.

EFFECT OF PRESSURE

As with concentration, a change in pressure, at constant temperature, on an equilibrium system containing gases has no effect on the value of either K_c or K_p.

Once again, any change in the equilibrium position occurs in order to maintain the equilibrium constant at a constant value.

If the partial pressure of only *one* of the component gases is changed, then the overall effect on K_p can be predicted in the same way the effects of changes in concentration on K_c were described in the previous section.

However, if the *total* pressure of a gaseous system is suddenly increased or decreased, then the partial pressures of *all* the gases will either increase or decrease. The effect on the position of equilibrium can be explained using the equilibrium law. This is shown in the following worked example.

WORKED EXAMPLE 3

Consider the reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

for which:

$$K_p = \frac{(p_{NH_3})^2}{(p_{N_2})(p_{H_2})^3}$$

Suppose the equilibrium partial pressures of nitrogen, hydrogen and ammonia are a, b and c atm, respectively. Then:

$$K_p = \frac{c^2}{ab^3}$$

If the total pressure is suddenly doubled, the partial pressures of nitrogen, hydrogen and ammonia will be doubled to $2a$, $2b$ and $2c$ respectively:

$$Q_p = \frac{(2c)^2}{2a \times (2b)^3} = \frac{4c^2}{16ab^3}$$

The value of Q_p is now one-quarter of the value of K_p. In order to re-establish equilibrium the numerator has to increase, with a subsequent decrease in the denominator, until $Q_p = K_p$. This is achieved by some nitrogen and hydrogen reacting to form more ammonia.

Hence, an increase in pressure increases the equilibrium yield of ammonia by shifting the equilibrium to the right.

EFFECT OF ADDING A CATALYST

The expression for an equilibrium constant of a given reaction contains only those substances included in the overall stoichiometric equation for the reaction. A catalyst does *not* appear in the overall equation, and so cannot influence the value of the equilibrium constant, and hence the equilibrium position.

We can think about this in another way. In **Topic 9 (Book 1: IAS)** you learned that a catalyst increases the rate of both the forward reaction and the backward reaction to the same extent. The catalyst therefore increases the rate at which equilibrium is established, but has no effect on the final concentrations of reactants and products at equilibrium. Therefore, a catalyst has no effect on the value of the equilibrium constant for a reaction.

CHECKPOINT

1. A dynamic equilibrium is set up between carbon monoxide, hydrogen and methanol. The equation for the reaction is:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

State the effect that an increase in pressure, at constant temperature, has on the value of K_c and also on the position of equilibrium. Justify your answers.

SKILLS CREATIVITY

2. The reaction between ethanoic acid and ethanol is reversible. The equation for the reaction is:

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

Some ethanoic acid and ethanol are mixed together, and some sulfuric acid is added to act as a catalyst. The mixture is left for several days to reach equilibrium.

(a) Explain what happens to the composition of the equilibrium mixture if some more ethanol is added.

(b) Explain what happens to the composition of the equilibrium mixture if some more sulfuric acid is added.

3. The reaction between hydrogen and iodine to form hydrogen iodide is reversible:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Some $H_2(g)$ is mixed with $I_2(g)$ and the mixture is allowed to reach equilibrium. At equilibrium, 0.14 mol of $H_2(g)$ is present.

The mixture was compressed to reduce its volume and then left to reach equilibrium at the original temperature. Explain the change, if any, in the composition of the equilibrium mixture.

SUBJECT VOCABULARY

reaction quotient a measure of the relative amounts of products and reactants present during a reaction at any given time

13A 5 RELATING ENTROPY TO EQUILIBRIUM CONSTANTS

SPECIFICATION REFERENCE 13.8 13.9

LEARNING OBJECTIVES

- Understand the effect of a change in temperature on
  (i) the value of ΔS_{total}
  (ii) the magnitude of the equilibrium constant, since $\Delta S_{total} = R \ln K$
- Be able to apply knowledge of the value of equilibrium constants to predict the extent to which a reaction takes place.

EFFECT OF TEMPERATURE CHANGE ON THE VALUE OF ΔS_{total}

We are going to use ideas from **Topic 12** on entropy and apply them to equilibrium situations.

In **Topic 12**, we introduced the equation for calculating the total entropy change:

$$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$$

We also stated that ΔS_{total} is positive for all spontaneous changes.

There is very little change in ΔS_{system} with a change in temperature, unless there is a change in state of one of the reactants or products. However, there are significant changes to $\Delta S_{surroundings}$.

The entropy change of the surroundings during a chemical reaction is given by:

$$\Delta S_{surroundings} = -\frac{\Delta H}{T}$$

where ΔH is the enthalpy change and T is the absolute temperature (i.e. the temperature measured in kelvin).

We can use this information to find out if a reaction is spontaneous at a particular temperature.

WORKED EXAMPLE 1

We will look at the decomposition of calcium carbonate into calcium oxide and carbon dioxide:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \qquad \Delta H = +177.9\ kJ\ mol^{-1}$$

$$\Delta S_{system} = +160.4\ J\ K^{-1}\ mol^{-1}$$

At 293 K (20 °C), $\Delta S_{surroundings} = -\dfrac{+177\,900\ J\ mol^{-1}}{293\ K} = -607.2\ J\ K^{-1}\ mol^{-1}$.

$$\Delta S_{total}(293\ K) = (+160.4 - 607.2) = -446.8\ J\ K^{-1}\ mol^{-1}$$

At 1173 K (900 °C), $\Delta S_{surroundings} = +177\,900\ J\ mol^{-1}/1173\ K$

$$= -151.7\ J\ K^{-1}\ mol^{-1}$$

$$\Delta S_{total}(1173\ K) = (+160.4 - 151.7) = +8.7\ J\ K^{-1}\ mol^{-1}$$

So the decomposition of calcium carbonate is not spontaneous at 293 K, but it is when heated to 1173 K.



▲ **fig A** Speed skaters can go fast because the pressure of the skates on the ice is changing the equilibrium position between water as a solid and water as a liquid.

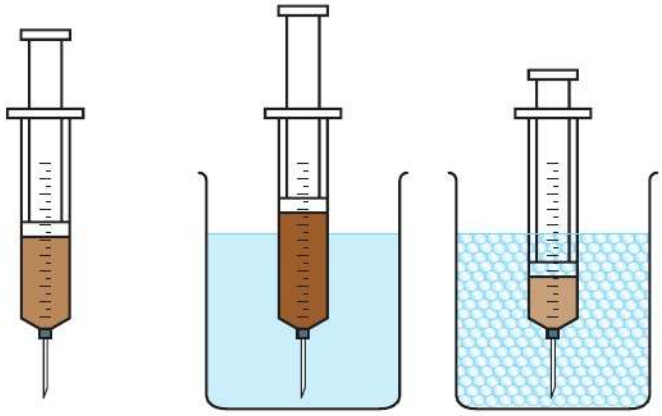
RELATIONSHIP BETWEEN TOTAL ENTROPY CHANGE, ΔS_{total}, AND THE EQUILIBRIUM CONSTANT, K

For a reversible reaction that can reach equilibrium, the equilibrium position can be reached from either side of the reaction. This means that both the forward and the backward reactions are spontaneous, so ΔS_{total} must be positive in both directions.

You may think this is impossible. To explain it we can look at the equilibrium reaction

$$N_2O_4(g) \rightleftharpoons 2\ NO_2(g)$$

colourless brown



(a) room temperature (b) hot water (c) crushed ice

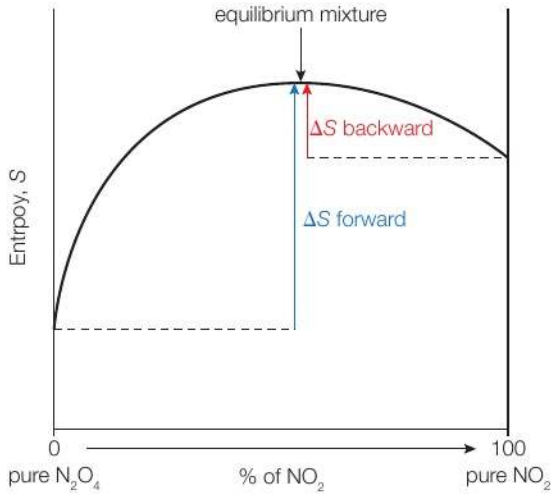
▲ **fig B** A mixture of N_2O_4 gas and NO_2 gas in a gas syringe: (a) at room temperature; (b) immersed in hot water; (c) immersed in crushed ice.

The differences in colour in **fig B** show how the position of equilibrium changes as the temperature changes.

In hot water, the mixture becomes darker brown. The equilibrium position has shifted to the right to produce more NO_2. Note that the volume has become larger because the mixture expands as it gets hotter.

In iced water, the mixture becomes lighter brown. This shows that the equilibrium position has shifted to the left to produce more N_2O_4. Note that the volume has become smaller because the reaction mixture has cooled.

Fig C shows how it is possible for the reaction to be spontaneous in each direction.



▲ **fig C** A graph of entropy against percentage NO_2 in a mixture of N_2O_4 and NO_2.

The entropy of N_2O_4 is less than the entropy of the equilibrium mixture. The change in entropy from pure N_2O_4 to the equilibrium mixture is positive, so the change is spontaneous.

Similarly, the change in entropy from NO_2 to the equilibrium mixture is also positive.

The entropy change for a mixture of the two gases in any proportions moving to the equilibrium mixture is also positive, so such changes are spontaneous.

Fig C also shows that neither the forward reaction nor the backward reaction can go to completion, as the entropy change from the equilibrium mixture to either reactant is negative.

At equilibrium, the total entropy change is zero and

$$\Delta S_{\text{total}}\,[\text{forward reaction}] = \Delta S_{\text{total}}\,[\text{backward reaction}]$$

The relationship between the total entropy of the reaction and the equilibrium constant is given by

$$\Delta S_{\text{total}} = R\ln K$$

where K can be K_c or K_p, and R is the gas constant.

Note that this matches with the conclusion that a high value of K corresponds to a large positive total entropy change.

USING TOTAL ENTROPY CHANGE TO CALCULATE AN EQUILIBRIUM CONSTANT

As mentioned above, $\Delta S_{\text{total}} = R\ln K$.

Rearranging the equation gives us:

$$\ln K = \frac{\Delta S_{\text{total}}}{R}$$

So:

$$K = e^{\Delta S_{\text{total}}/R}$$

LEARNING TIP

The thermodynamic value of K calculated from $\Delta S^{\ominus}_{\text{total}} = R\ln K$ has no units. This is because you can take the logarithm of a number but not of a quantity with units. The value of K calculated from experimental data does have units.

To convert an experimental value of K_p into a thermodynamic value, you need to remove the units by dividing each partial pressure by the standard pressure of 1 atm. Since each partial pressure has been divided by a factor of 1, the numerical value does not alter.

The same is true for K_c. To convert an experimental value of K_c into a thermodynamic value, you need to remove the units by dividing each concentration by the standard concentration of 1 mol dm^{-3}.

WORKED EXAMPLE 2

The equation for the reaction to form sulfur trioxide from sulfur dioxide and oxygen is:

$$SO_2(g) + \tfrac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \qquad \Delta S^{\ominus}_{total} = 238.3\ J\ K^{-1}\ mol^{-1}$$

Calculate a value for the equilibrium constant, K, for this reaction ($R = 8.31\ J\ mol^{-1}\ K^{-1}$).

$$K = e^{\Delta S_{total}/R}$$
$$= e^{238.3/8.31}$$
$$= 2.84 \times 10^{12}$$

To solve this calculation, press the e^x or $e^{\square}$ button on your calculator. Then enter 238.3 ÷ 8.31 followed by the equals button. The answer will be displayed.

LEARNING TIP

Remember that a value of K calculated from thermodynamic data has no units.

RELATIONSHIP BETWEEN EQUILIBRIUM CONSTANT AND POSITION OF EQUILIBRIUM

There is no firm rule about the relationship between the equilibrium constant, K, and the position of equilibrium of the reaction.

However, as a general rule we can say that a very large value of K suggests that the position of equilibrium is well over to the products side (i.e. the right-hand side of the equation).

Similarly, a very small value of K suggests that the position of equilibrium is well over to the reactants side (i.e. the left-hand side of the equation).

CHECKPOINT

1. Hydrogen can be manufactured by reacting methane with steam. The equation for the reaction is:

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g); \qquad \Delta H_r^{\ominus} = +206\ kJ\ mol^{-1}$$

(a) The equilibrium constant, K_p, for this reaction, at a given temperature, has a numerical value of 8.54.

Calculate the total entropy change at the temperature of the reaction.

(b) Use your answer to (a) to calculate the temperature at which this reaction reaches equilibrium.

($\Delta S^{\ominus}_{system} = +225\ J\ K^{-1}\ mol^{-1}$; $R = 8.31\ J\ mol^{-1}\ K^{-1}$)

(c) Use the magnitude and signs of the entropy changes to explain the effect of a temperature increase on the equilibrium constant for this reaction.

2. This question is about the equilibrium reaction:

$$Fe^{2+}(aq) + Ag^{+}(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$$

The equilibrium is reached slowly.

(a) State two observations that are made if aqueous solutions containing iron(II) ions and silver ions are mixed and allowed to stand for several hours.

(b) (i) $\Delta S^{\ominus}_{total}$ for this reaction is + 47.64 J K^{-1} mol^{-1}.

Calculate the numerical value for the equilibrium constant, K_c.

($R = 8.31\ J\ mol^{-1}\ K^{-1}$)

(ii) $\Delta S^{\ominus}_{system}$ for this reaction is −208.3 J K^{-1} mol^{-1}.

Explain why this value is negative.

(iii) Calculate $\Delta S^{\ominus}_{surroundings}$ for this reaction.

(iv) Use your answer to (b)(iii) to calculate $\Delta H^{\ominus}$ for this reaction, and then explain the effect of increasing the temperature on the value of ΔS_{total}.

SKILLS ADAPTIVE LEARNING